

LARGE NUMBERS AROUND US



0774CH01

1.1 A Lakh Varieties!

Eshwarappa is a farmer in Chintamani, a town in Karnataka. He visits the market regularly to buy seeds for his rice field. During one such visit he overheard a conversation between Ramanna and Lakshamma. Ramanna said, “Earlier our country had about a lakh varieties of rice. Farmers used to preserve different varieties of seeds and use them to grow rice. Now, we only have a handful of varieties. Also, farmers have to come to the market to buy seeds”.



Lakshamma said, “There is a seed bank near my house. So far, they have collected about a hundred indigenous varieties of rice seeds from different places. You can also buy seeds from there.”

You may have heard the word ‘lakh’ before. Do you know how big one lakh is? Let us find out.

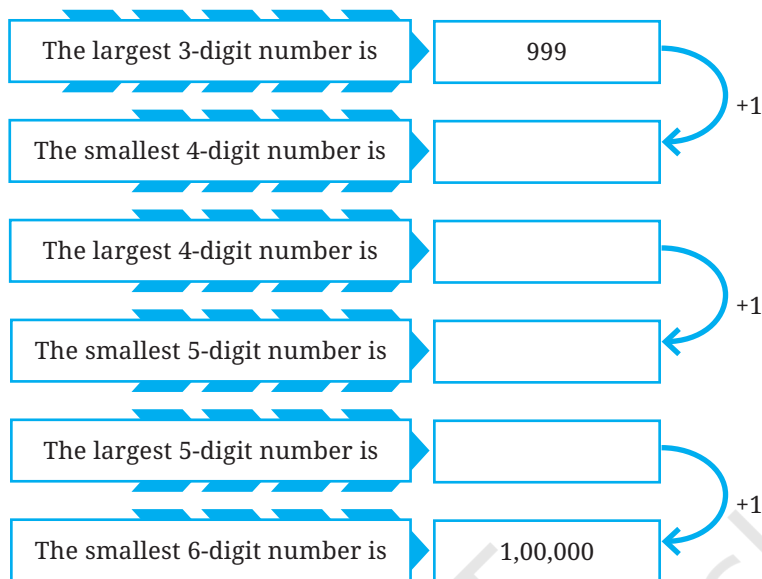
Eshwarappa shared this incident with his daughter Roxie and son Estu .

Estu was surprised to know that there were about one lakh varieties of rice in this country. He wondered “One lakh! So far I have only tasted 3 varieties. If we tried a new variety each day, would we even come close to tasting all the varieties in a lifetime of 100 years?”

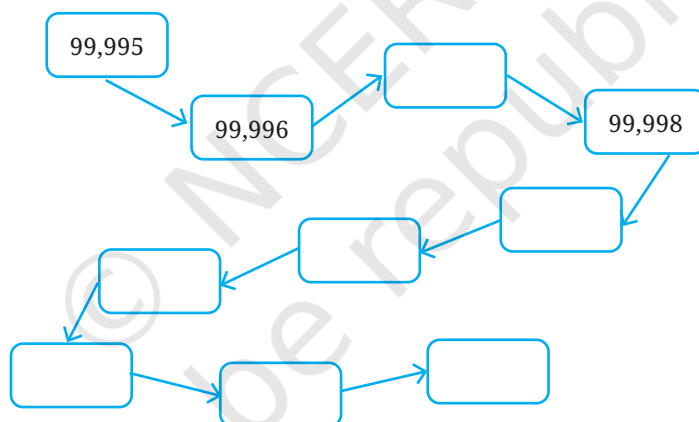
What do you think? Guess.



But how much is one lakh? Observe the pattern and fill in the boxes given below.



1,00,000 is read as "One Lakh"



Roxie and Estu found that if they ate one variety of rice a day, they would come nowhere close to a lakh in a lifetime! Roxie suggests, "What if we ate 2 varieties of rice every day? Would we then be able to eat 1 lakh varieties of rice in 100 years?"



- ? What if a person ate 3 varieties of rice every day? Will they be able to taste all the lakh varieties in a 100 year lifetime? Find out.

Estu said, "We know how many days there are in a year—365, if we ignore leap years. If we live for y years, the number of days in our lifetime will be $365 \times y$."

- ❓ Choose a number for y . How close to one lakh is the number of days in y years, for the y of your choice?

❓ **Figure it Out**

1. According to the 2011 Census, the population of the town of *Chintamani* was about 75,000. How much less than one lakh is 75,000?
2. The estimated population of *Chintamani* in the year 2024 is 1,06,000. How much more than one lakh is 1,06,000?
3. By how much did the population of *Chintamani* increase from 2011 to 2024?

Getting a Feel of Large Numbers

You may have come across interesting facts like these:

- The world's tallest statue is the 'Statue of Unity' in Gujarat depicting Sardar Vallabhbhai Patel. Its height is about 180 metres.
- Kunchikal waterfall in Karnataka is said to drop from a height of about 450 metres.

It is not always easy to get a sense of how big these measurements are. But, we can get a better sense of their size when we compare them with something familiar. Let us see an example.

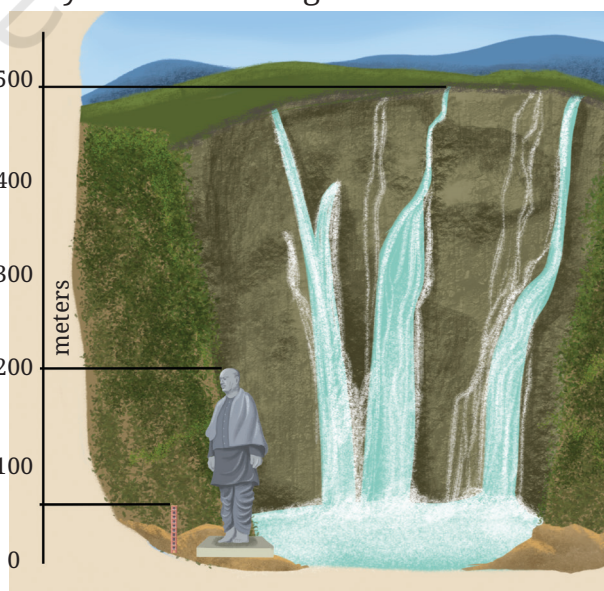
Look at the picture on the right. Somu is 1 metre tall. If each floor is about four times his height, what is the approximate height of the building?

- ❓ Which is taller — The Statue of Unity or this building? How much taller?
_____ m.

We can see that the height of the Statue of Unity is close to 4 times the height of Somu's building.

- ❓ How much taller is the Kunchikal waterfall than Somu's building?
_____ m.

- ❓ How many floors should Somu's building have to be as high as the waterfall?
_____ .



Is One Lakh a Very Large Number?

Eshwarappa asked Roxie and Estu, “Is a lakh big or small?”

Roxie feels that 1 lakh is a large number:

- “We had one lakh varieties of rice—that is a lot.”
- “Living 1 lakh days would mean living for 274 years—that is a really long time!”
- “If 1 lakh people stood shoulder to shoulder in a line they could stretch as far as 38 kilometres.”



Estu, however, thinks it is not that big:

- “Do you know that the cricket stadium in Ahmedabad has a seating capacity of more than 1 lakh? One lakh people in such a small area!”
- “Most humans have 80,000 to 1,20,000 hairs on their heads. Imagine, 1 lakh hairs fit in such a tiny space!”
- “I heard that there are some species of fish where a female fish can lay almost one lakh eggs at once very comfortably. Some even lay tens of lakhs of eggs at a time.”



? How do you view a lakh—is a lakh big or small?

Reading and Writing Numbers

We have already been using commas for 5-digit numbers like 45,830 in the Indian place value system. As numbers grow bigger, using commas helps in reading the numbers easily. We use a comma in between the digits representing the “ten thousands” place and the “one lakh” place as you have seen just before (1,00,000).

The number name of 12,78,830 is twelve lakh seventy eight thousand eight hundred thirty.

Similarly, the number 15,75,000 in words is fifteen lakh seventy five thousand.

Write each of the numbers given below in words:

- 3,00,600
- 5,04,085
- 27,30,000
- 70,53,138

Write the corresponding number in the Indian place value system for each of the following:

- One lakh twenty three thousand four hundred and fifty six
- Four lakh seven thousand seven hundred and four
- Fifty lakhs five thousand and fifty
- Ten lakhs two hundred and thirty five

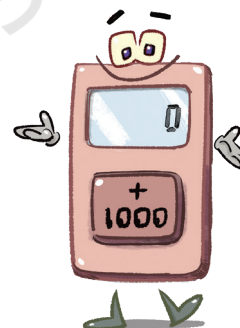
Note to the Teacher: Encourage students to make connections between these facts. For example, can the whole population of Chintamani fit in the stadium? How can we imagine the line of 38 km, having a lakh people, sitting next to each other in the stadium?

1.2 Land of Tens

In the **Land of Tens**, there are special calculators with special buttons.

- The **Thoughtful Thousands** only has a +1000 button. How many times should it be pressed to show:

- Three thousand? 3 times
- 10,000? _____
- Fifty three thousand? _____
- 90,000? _____
- One Lakh? _____
- _____ ? 153 times
- How many thousands are required to make one lakh?



- The **Tedious Tens** only has a +10 button. How many times should it be pressed to show:

- Five hundred? _____
- 780? _____
- 1000? _____
- 3700? _____
- 10,000? _____
- One lakh? _____
- _____ ? 435 times



- The **Handy Hundreds** only has a +100 button. How many times should it be pressed to show:

- Four hundred? _____ times
- 3,700? _____



- (c) 10,000? _____
- (d) Fifty three thousand? _____
- (e) 90,000? _____
- (f) 97,600? _____
- (g) 1,00,000? _____
- (h) _____? 582 times
- (i) How many hundreds are required to make ten thousand?
- (j) How many hundreds are required to make one lakh?
- (k) Handy Hundreds says, "There are some numbers which Tedious Tens and Thoughtful Thousands can't show but I can." Is this statement true? Think and explore.

4. **Creative Chitti** is a different kind of calculator. It has the following buttons: +1, +10, +100, +1000, +10000, +100000 and +1000000. It always has multiple ways of doing things. "How so?", you might ask. To get the number 321, it presses +10 thirty two times and +1 once. Will it get 321? Alternatively, it can press +100 two times and +10 twelve times and +1 once.



5. Two of the many different ways to get 5072 are shown below:

These two ways can be expressed as:

(a) $(50 \times 100) + (7 \times 10) + (2 \times 1) = 5072$

(b) $(3 \times 1000) + (20 \times 100) + (72 \times 1) = 5072$

- ?** Find a different way to get 5072 and write an expression for the same.

? **Figure it Out**

For each number given below, write expressions for at least two different ways to obtain the number through button clicks. Think like Chitti and be creative.

- (a) 8300
- (b) 40629
- (c) 56354

Buttons	5072	
+10,00,000		
+1,00,000		
+10,000		
+1,000		3
+100	50	20
+10	7	
+1	2	72

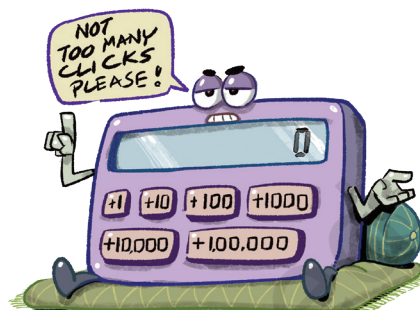
- (d) 66666
(e) 367813

Creative Chitti has some questions for you —

- (a) You have to make exactly 30 button presses. What is the largest 3-digit number you can make? What is the smallest 3-digit number you can make?
- (b) 997 can be made using 25 clicks. Can you make 997 with a different number of clicks?

Create questions like these and challenge your classmates.

Systematic Sippy is a different kind of calculator. It has the following buttons: +1, +10, +100, +1000, +10000, +100000. It wants to be used as minimally as possible.



- ? How can we get the numbers (a) 5072, (b) 8300 using as few button clicks as possible?

Find out which buttons should be clicked and how many times to get the desired numbers given in the table. The aim is to click as few buttons as possible.

Here is one way to get the number 5072. This method uses 23 button clicks in total.

Is there another way to get 5072 using less than 23 button clicks? Write the expression for the same.

Buttons	5072
+10,00,000	
+1,00,000	
+10,000	
+1,000	5
+100	0
+10	6
+1	12

? **Figure it Out**

- For the numbers in the previous exercise, find out how to get each number by making the smallest number of button clicks and write the expression.
- Do you see any connection between each number and the corresponding smallest number of button clicks?
- If you notice, the expressions for the least button clicks also give the Indian place value notation of the numbers. Think about why this is so.



What if we press the +10,00,000 button ten times? What number will come up? How many zeroes will it have? What should we call it? The number will be 100 lakhs, which is also called a **crore**. 1 crore is written as 1,00,00,000 — it is 1 followed by seven zeroes.

1.3 Of Crores and Crores!

The table below shows some numbers according to both the Indian system and the American system (also called the International system) of naming numerals and placing commas. Observe the placement of commas in both systems.

Indian System		American System	
1,000	One thousand	1,000	One thousand
10,000	Ten thousand	10,000	Ten thousand
1,00,000	One lakh	100,000	Hundred thousand
10,00,000	Ten lakhs	1,000,000	One million
1,00,00,000	One crore	10,000,000	Ten million
10,00,00,000	Ten crores	100,000,000	Hundred million
1,00,00,00,000	One arab or One hundred crores	1,000,000,000	One billion

Notice that in the Indian system, commas are placed to group the digits in a 3-2-2-2... pattern from right to left (thousands, lakhs, crores, etc.). In the American system, the digits are grouped uniformly in a 3-3-3-3... pattern from right to left (thousands, millions, billions, etc.).

The Indian system of naming numbers is also followed in Bhutan, Nepal, Sri Lanka, Pakistan, Bangladesh, Maldives, Afghanistan, and Myanmar. The words lakh and crore originate from the Sanskrit words *lakṣha* (लक्ष) and *koṭi* (कोटि). The American system is also used in many countries.

Observe the number of zeroes in 1 lakh and 1 crore.

1 lakh, written in numbers would be 1 followed by 5 zeroes.

1 crore, written in numbers would be 1 followed by 7 zeroes.

A lakh is a hundred times a thousand, a crore is a hundred times a lakh and an arab is a hundred times a crore (i.e., a hundred thousand is 1 lakh, 100 lakhs is 1 crore, and 100 crores is 1 arab).

? How many zeros does a thousand lakh have? ____

- ? How many zeros does a hundred thousand have? ____

The number 9876501234 can be easily read by placing commas first:

- (a) 9,87,65,01,234 → 9 arab 87 crore 65 lakhs 1 thousand and 234 or 987 crore 65 lakh 1 thousand 234 (in the Indian system).
 (b) 9,876,501,234 → 9 billion 876 million 501 thousand 234 (in the American system).

? **Figure it Out**

- Read the following numbers in Indian place value notation and write their number names in both the Indian and American systems:
 - 4050678
 - 48121620
 - 20022002
 - 246813579
 - 345000543
 - 1020304050
- Write the following numbers in Indian place value notation:
 - One crore one lakh one thousand ten
 - One billion one million one thousand one
 - Ten crore twenty lakh thirty thousand forty
 - Nine billion eighty million seven hundred thousand six hundred
- Compare and write '<', '>' or '=':
 - 30 thousand ____ 3 lakhs
 - 500 lakhs ____ 5 million
 - 800 thousand ____ 8 million
 - 640 crore ____ 60 billion

We shall come across even bigger numbers in later grades.

1.4 Exact and Approximate Values



What do you think of this conversation? Have you read or heard such headlines or statements?

Very often, exact numbers are not required and just an approximation is sufficient. For example, according to the 2011 census, the population of Chintamani town is 76,068. Instead, saying that the population is about 75,000 is enough to give an idea of how big the quantity is.



There are situations where it makes sense to **round up** a number (rounding up is when the approximated number is more than the actual number). For example, if a school has 732 people including students, teachers and staff: the principal might order 750 sweets instead of 700 sweets.

There are situations where it is better to **round down** (rounding down is when the approximated number is less than the actual number). For example, if the cost of an item is ₹470, the shopkeeper may say that the cost is around ₹450 instead of saying it is around ₹500.

- ? Think and share situations where it is appropriate to (a) round up, (b) round down, (c) either rounding up or rounding down is okay and (d) when exact numbers are needed.

Nearest Neighbours

With large numbers it is useful to know the nearest thousand, lakh or crore. For example, the nearest neighbours of the number 6,72,85,183 are shown in the table below.

Nearest thousand	6,72,85,000
Nearest ten thousand	6,72,90,000
Nearest lakh	6,73,00,000
Nearest ten lakh	6,70,00,000
Nearest crore	7,00,00,000

❓ Similarly, write the five nearest neighbours for these numbers:

(a) 3,87,69,957

(b) 29,05,32,481

❓ I have a number for which all five nearest neighbours are 5,00,00,000. What could the number be? How many such numbers are there?



Roxie and Estu are estimating the values of simple expressions.

1. $4,63,128 + 4,19,682$,

Roxie: "The sum is near 8,00,000 and is more than 8,00,000."

Estu: "The sum is near 9,00,000 and is less than 9,00,000."

(a) Are these estimates correct? Whose estimate is closer to the sum?

(b) Will the sum be greater than 8,50,000 or less than 8,50,000? Why do you think so?

(c) Will the sum be greater than 8,83,128 or less than 8,83,128? Why do you think so?

(d) Exact value of $4,63,128 + 4,19,682 =$ _____

2. $14,63,128 - 4,90,020$

Roxie: "The difference is near 10,00,000 and is less than 10,00,000."

Estu: "The difference is near 9,00,000 and is more than 9,00,000."

(a) Are these estimates correct? Whose estimate is closer to the difference?

(b) Will the difference be greater than 9,50,000 or less than 9,50,000? Why do you think so?

(c) Will the difference be greater than 9,63,128 or less than 9,63,128? Why do you think so?

(d) Exact value of $14,63,128 - 4,90,020 =$ _____

Note to the Teacher: Ask students questions like — “what could the numbers be if the sum had to be less than 8,50,000.”

Populations of Cities

Observe the populations of some Indian cities in the **table below**.

Rank	City	Population (2011)	Population (2001)
1	Mumbai	1,24,42,373	1,19,78,450
2	New Delhi	1,10,07,835	98,79,172
3	Bengaluru	84,25,970	43,01,326
4	Hyderabad	68,09,970	36,37,483
5	Ahmedabad	55,70,585	35,20,085
6	Chennai	46,81,087	43,43,645
7	Kolkata	44,86,679	45,72,876
8	Surat	44,67,797	24,33,835
9	Vadodara	35,52,371	16,90,000
10	Pune	31,15,431	25,38,473
11	Jaipur	30,46,163	23,22,575
12	Lucknow	28,15,601	21,85,927
13	Kanpur	27,67,031	25,51,337
14	Nagpur	24,05,665	20,52,066

15	Indore	19,60,631	14,74,968
16	Thane	18,18,872	12,62,551
17	Bhopal	17,98,218	14,37,354
18	Visakhapatnam	17,28,128	13,45,938
19	Pimpri-Chinchwad	17,27,692	10,12,472
20	Patna	16,84,222	13,66,444

 From the information given in the table, answer the following questions by approximation:

1. What is your general observation about this data? Share it with the class.
2. What is an appropriate title for the above table?
3. How much is the population of Pune in 2011? Approximately, by how much has it increased compared to 2001?
4. Which city's population increased the most between 2001 and 2011?
5. Are there cities whose population has almost doubled? Which are they?
6. By what number should we multiply Patna's population to get a number/population close to that of Mumbai?

1.5 Patterns in Products

Roxie and Estu are playing with multiplication. They encounter an interesting technique for multiplying a number by 10, 100, 1000, and so on.

A Multiplication Shortcut

Roxie evaluated 116×5 as follows:

$$\begin{aligned}
 116 \times 5 &= \overset{58}{\cancel{116}} \times \frac{10}{\cancel{2}} \\
 &= 58 \times 10 \\
 &= 580.
 \end{aligned}$$

Estu evaluated 824×25 as follows:

$$\begin{array}{rcl} 824 \times 25 & = & \begin{array}{r} 206 \\ 824 \times \frac{100}{4} \\ \hline \end{array} \\ & = & 20600. \end{array}$$

- ?** Using the meaning of multiplication and division, can you explain why multiplying by 5 is the same as dividing by 2 and multiplying by 10?



? **Figure it Out**

1. Find quick ways to calculate these products:

(a) $2 \times 1768 \times 50$

(b) 72×125 [**Hint:** $125 = \frac{1000}{8}$]

(c) $125 \times 40 \times 8 \times 25$

2. Calculate these products quickly.

(a) $25 \times 12 =$ _____

(b) $25 \times 240 =$ _____

(c) $250 \times 120 =$ _____

(d) $2500 \times 12 =$ _____

(e) _____ $\times$ _____ $= 120000000$

How Long is the Product?

In each of the following boxes, the multiplications produce interesting patterns. Evaluate them to find the pattern. Extend the multiplications based on the observed pattern.

$$\begin{array}{l} 11 \times 11 = \\ 111 \times 111 = \\ 1111 \times 1111 = \end{array}$$

$$\begin{array}{l} 66 \times 61 = \\ 666 \times 661 = \\ 6666 \times 6661 = \end{array}$$

$$\begin{array}{l} 3 \times 5 = \\ 33 \times 35 = \\ 333 \times 335 = \end{array}$$

$$\begin{array}{l} 101 \times 101 = \\ 102 \times 102 = \\ 103 \times 103 = \end{array}$$

- ❓ Observe the number of digits in the two numbers being multiplied and their product in each case. Is there any connection between the numbers being multiplied and the number of digits in their product?
- ❓ Roxie says that the product of two 2-digit numbers can only be a 3- or a 4-digit number. Is she correct?
- ❓ Should we try all possible multiplications with 2-digit numbers to tell whether Roxie's claim is true? Or is there a better way to find out?



She explains her reasoning: "We want to know about the number of digits in the product of two 2-digit numbers. To know the smallest such product I took 10×10 , so all other products will be greater than 100.

To know the greatest such product I multiplied the smallest 3-digit numbers (100×100) to get 10,000; so the product of all the 2-digit multiplications will be less than 10,000."

- ❓ Can multiplying a 3-digit number with another 3-digit number give a 4-digit number?
- ❓ Can multiplying a 4-digit number with a 2-digit number give a 5-digit number?
- ❓ Observe the multiplication statements below. Do you notice any patterns? See if this pattern extends for other numbers as well.

1-digit	×	1-digit	=	1-digit	or	2-digit
2-digit	×	1-digit	=	2-digit	or	3-digit
2-digit	×	2-digit	=	3-digit	or	4-digit
3-digit	×	3-digit	=	5-digit	or	6-digit
5-digit	×	5-digit	=		or	
8-digit	×	3-digit	=		or	
12-digit	×	13-digit	=		or	

Fascinating Facts about Large Numbers

Some interesting facts about large numbers are hidden below. Calculate the product or quotient to uncover the facts. Once you find the product or quotient, read the number in both Indian and American naming systems. Share your thoughts and questions about the fact with the class after you discover each number.

$$1250 \times 380$$

is the number of *kīrtanas* composed by Purandaradāsa according to legends.

Purandaradāsa was a composer and singer in the 15th century. His *kīrtanas* spanned social reform, *bhakti* and spirituality. He systematised methods for teaching Carnatic music which is followed to the present day.



How many years did he live to compose so many songs? At what age did he start composing songs?

If he composed 4,75,000 songs, how many songs per year did he have to compose?

$$2100 \times 70,000$$

is the approximate distance in kilometers, between the Earth and the Sun.

This distance keeps varying throughout the year. The farthest distance is about 152 million kilometers.



How did they measure
the distance between the
Earth and the Sun?

$$6400 \times 62,500$$

is the average number of litres of water the Amazon river discharges into the Atlantic Ocean every second.

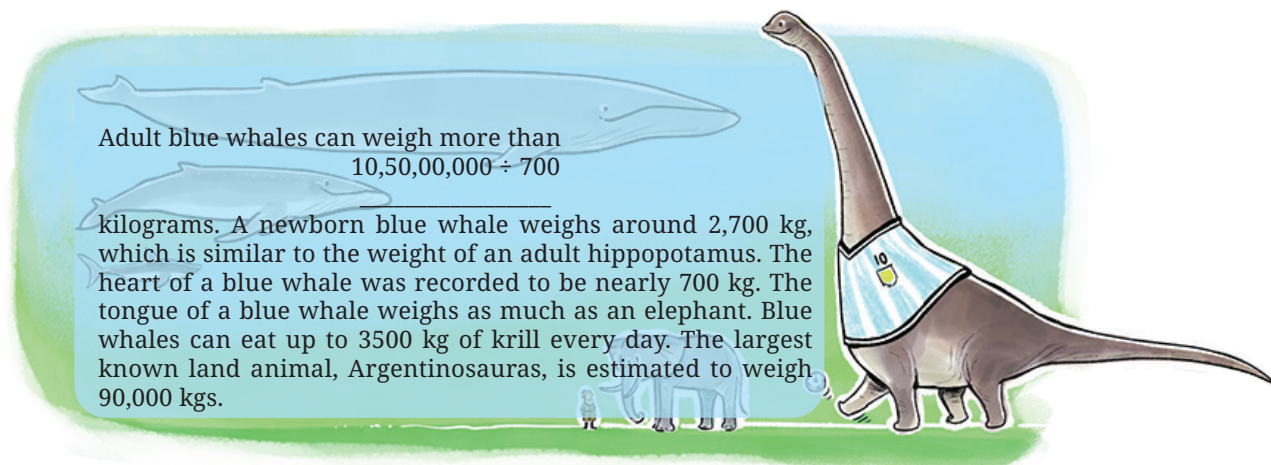
The river's flow into the Atlantic is so much that drinkable freshwater is found even 160 kilometers into the open sea.

As you did before, divide the given numbers to uncover interesting facts about division. Share your thoughts and questions with the class after you uncover each number.

$$13,95,000 \div 150$$

is the distance (in kms) of the longest single-train journey in the world. The train runs in Russia between Moscow and Vladivostok. The duration of this journey is about 7 days. The longest train route in India is from Dibrugarh in Assam to Kanyakumari in Tamil Nadu; it covers 4219 kms in about 76 hours.





Large Number Fact

In a single gram of healthy soil there can be 100 million to 1 billion bacteria and 1 lakh to 1 million fungi, which can support plants' growth and health.

Share such large-number facts you know / come across with your class.

1.6 Did You Ever Wonder...?

Estu is amused by all these interesting facts about large numbers. While thinking about these, he came up with an unusual question, "Could the entire population of Mumbai fit into 1 lakh buses?"

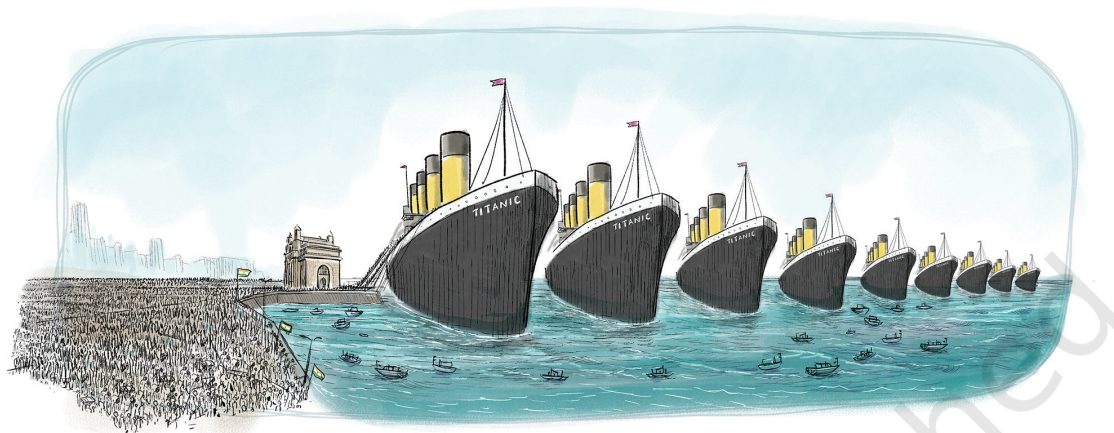
What do you think?

How can we find out?

Let us assume a bus can accommodate 50 people. Then 1 lakh buses can accommodate $1 \text{ lakh} \times 50 = 50 \text{ lakh}$ people.

The population of Mumbai is more than 1 crore 24 lakhs (we saw this in an earlier table). So, the entire population of Mumbai cannot fit in 1 lakh buses.

- ? The RMS Titanic ship carried about 2500 passengers. Can the population of Mumbai fit into 5000 such ships?



Inspired by this strange question, Roxie wondered, “If I could travel 100 kilometers every day, could I reach the Moon in 10 years?” (The distance between the Earth and the Moon is 3,84,400 km.)

How far would she have travelled in a year?

How far would she have travelled in 10 years?

Is it not easier to perform these calculations in stages? You can use this method for all large calculations.

- ? Find out if you can reach the Sun in a lifetime, if you travel 1000 kilometers every day. (You had written down the distance between the Earth and the Sun in a previous exercise.)
- ? Make necessary reasonable assumptions and answer the questions below:
- (a) If a single sheet of paper weighs 5 grams, could you lift one lakh sheets of paper together at the same time?
 - (b) If 250 babies are born every minute across the world, will a million babies be born in a day?
 - (c) Can you count 1 million coins in a day? Assume you can count 1 coin every second.
- ? Think and create more such fun questions and share them with your class.

? **Figure it Out**

1. Using all digits from 0–9 exactly once (the first digit cannot be 0) to create a 10-digit number, write the —

- (a) Largest multiple of 5
 - (b) Smallest even number
2. The number 10,30,285 in words is Ten lakhs thirty thousand two hundred eighty five, which has 43 letters. Give a 7-digit number name which has the maximum number of letters.
 3. Write a 9-digit number where exchanging any two digits results in a bigger number. How many such numbers exist?
 4. Strike out 10 digits from the number 12345123451234512345 so that the remaining number is as large as possible.
 5. The words 'zero' and 'one' share letters 'e' and 'o'. The words 'one' and 'two' share a letter 'o', and the words 'two' and 'three' also share a letter 't'. How far do you have to count to find two consecutive numbers which do not share an English letter in common?
 6. Suppose you write down all the numbers 1, 2, 3, 4, ..., 9, 10, 11, ... The tenth digit you write is '1' and the eleventh digit is '0', as part of the number 10.

 - (a) What would the 1000th digit be? At which number would it occur?
 - (b) What number would contain the millionth digit?
 - (c) When would you have written the digit '5' for the 5000th time?
 7. A calculator has only '+10,000' and '+100' buttons. Write an expression describing the number of button clicks to be made for the following numbers:
 - (a) 20,800
 - (b) 92,100
 - (c) 1,20,500
 - (d) 65,30,000
 - (e) 70,25,700
 8. How many lakhs make a billion?
 9. You are given two sets of number cards numbered from 1 – 9. Place a number card in each box below to get the (a) largest possible sum (b) smallest possible difference of the two resulting numbers.

10. You are given some number cards; 4000, 13000, 300, 70000, 150000, 20, 5. Using the cards get as close as you can to the numbers below using any operation you want. Each card can be used only once for making a particular number.

- (a) 1,10,000: Closest I could make is $4000 \times (20 + 5) + 13000 = 1,13,000$
- (b) 2,00,000:
- (c) 5,80,000:
- (d) 12,45,000:
- (e) 20,90,800:

11. Find out how many coins should be stacked to match the height of the Statue of Unity. Assume each coin is 1 mm thick.

12. Grey-headed albatrosses have a roughly 7-feet wide wingspan. They are known to migrate across several oceans. Albatrosses can cover about 900 – 1000 km in a day. One of the longest single trips recorded is about 12,000 km. How many days would such a trip take to cross the Pacific Ocean approximately?

13. A bar-tailed godwit holds the record for the longest recorded non-stop flight. It travelled 13,560 km from Alaska to Australia without stopping. Its journey started on 13 October 2022 and continued for about 11 days. Find out the approximate distance it covered every day. Find out the approximate distance it covered every hour.

14. Bald eagles are known to fly as high as 4500 – 6000 m above the ground level. Mount Everest is about 8850 m high. Aeroplanes can fly as high as 10,000 – 12,800 m. How many times bigger are these heights compared to Somu's building?



SUMMARY

- We came across large numbers — lakhs, crores and arabs; millions and billions. We learnt how to read and write these numbers in the Indian and American/International naming systems.
 - (a) 1 lakh is 1 followed by 5 zeroes: 1,00,000
 - (b) 1 crore is 1 followed by 7 zeroes: 1,00,00,000
 - (c) 1 million is 1 followed by 6 zeroes: 1,000,000 (which is also ten lakhs)
 - (d) 1 arab is 1 followed by 9 zeroes: 1,000,000,000 (which is also 100 crore or 1 billion)
- We generally round up or round down large numbers. Many times it is enough just to know roughly how big or small something is.
- To get a sense of large numbers or quantities, we can check how many times bigger they are compared to numbers or quantities that are more familiar.
- We saw how to factorise numbers and regroup them to make multiplications simpler.
- We carried out interesting thought experiments such as — “Would one be able to watch 1000 movies in a year?”



IT'S PUZZLE TIME!

Toothpick Digits

We can write digits as shown in the image below:



You can either use toothpicks or matchsticks, or just write the digits in this way, using lines to represent sticks.

To make the digit 7, three sticks are needed.

Write or make the number 5108. How many sticks are required?

1. Make or write the number 42,019. It would require exactly 23 sticks.
2. Starting with 42,019, add or write two more sticks, and make a bigger number. One example is 42,078. What other numbers bigger than 42,019 can you make in this way?
3. Preetham wants to insert the digit '1' somewhere among the digits '4', '2', '0', '1' and '9'. Where should he place the digit '1' to get the biggest possible number?
4. What other numbers can he make by placing the digit '1'?

1. Make or write the number 63,890.
2. Starting with 63,890, rearrange exactly four sticks and make a bigger number. One example is 88,078. What other numbers bigger than 63,890 can you make in this way?

1. Make any number using exactly 24 sticks or lines.
2. What is the biggest number that can be made using 24 sticks or lines?
3. What is the smallest number that can be made using 24 sticks or lines?

Make your own questions and challenge each other.



2

ARITHMETIC EXPRESSIONS



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2.1 Simple Expressions

You may have seen mathematical phrases like $13 + 2$, $20 - 4$, 12×5 , and $18 \div 3$. Such phrases are called **arithmetic expressions**.

Every arithmetic expression has a value which is the number it evaluates to. For example, the value of the expression $13 + 2$ is 15. This expression can be read as ‘13 plus 2’ or ‘the sum of 13 and 2’.

We use the equality sign ‘=’ to denote the relationship between an arithmetic expression and its value. For example:

$$13 + 2 = 15.$$

- ❓ **Example 1:** Mallika spends ₹25 every day for lunch at school. Write the expression for the total amount she spends on lunch in a week from Monday to Friday.

The expression for the total amount is 5×25 .
 5×25 is “5 times 25” or “the product of 5 and 25”.

Different expressions can have the same value. Here are multiple ways to express the number 12, using two numbers and any of the four operations $+$, $-$, $\times$ and $\div$:

$$10 + 2, 15 - 3, 3 \times 4, 24 \div 2.$$

- ❓ Choose your favourite number and write as many expressions as you can having that value.

Comparing Expressions

As we compare numbers using ‘=’, ‘<’ and ‘>’ signs, we can also compare expressions. We compare expressions based on their values and write the ‘equal to’, ‘greater than’ or ‘less than’ sign accordingly. For example,

$$10 + 2 > 7 + 1$$

because the value of $10 + 2 = 12$ is greater than the value of $7 + 1 = 8$. Similarly,

$$13 - 2 < 4 \times 3.$$

? Figure it Out

- Fill in the blanks to make the expressions equal on both sides of the = sign:

(a) $13 + 4 = \underline{\quad} + 6$

(b) $22 + \underline{\quad} = 6 \times 5$

(c) $8 \times \underline{\quad} = 64 \div 2$

(d) $34 - \underline{\quad} = 25$

- Arrange the following expressions in ascending (increasing) order of their values.

(a) $67 - 19$

(b) $67 - 20$

(c) $35 + 25$

(d) 5×11

(e) $120 \div 3$

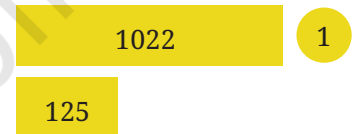
? Example 2: Which is greater? $1023 + 125$ or $1022 + 128$?

Imagining a situation could help us answer this without finding the values. Raja had 1023 marbles and got 125 more today. Now he has $1023 + 125$ marbles. Joy had 1022 marbles and got 128 more today. Now he has $1022 + 128$ marbles. Who has more?

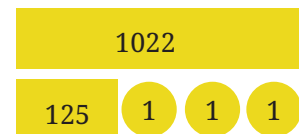
This situation can be represented as shown in the picture on the right. To begin with, Raja had 1 more marble than Joy. But Joy got 3 more marbles than Raja today. We can see that Joy has (two) more marbles than Raja now. That is,

$$1023 + 125 < 1022 + 128.$$

Raja ($1023 + 125$)



Joy ($1022 + 128$)



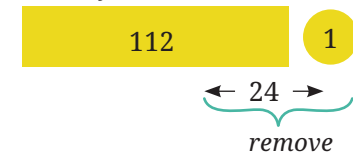
? Example 3: Which is greater? $113 - 25$ or $112 - 24$?

Imagine a situation, Raja had 113 marbles and lost 25 of them. He has $113 - 25$ marbles. Joy had 112 marbles and lost 24 today. He has $112 - 24$ marbles. Who has more marbles left with them?

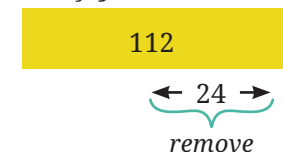
Raja had 1 marble more than Joy. But he also lost 1 marble more than Joy did. Therefore, they have an equal number of marbles now. That is,

$$113 - 25 = 112 - 24.$$

Raja ($113 - 25$)



Joy ($112 - 24$)



- ? Use '>' or '<' or '=' in each of the following expressions to compare them. Can you do it without complicated calculations? Explain your thinking in each case.

- (a) $245 + 289$ $246 + 285$
 (b) $273 - 145$ $272 - 144$
 (c) $364 + 587$ $363 + 589$
 (d) $124 + 245$ $129 + 245$
 (e) $213 - 77$ $214 - 76$

2.2 Reading and Evaluating Complex Expressions

Sometimes, when an expression is not accompanied by a context, there can be more than one way of evaluating its value. In such cases, we need some tools and rules to specify how exactly the expression has to be evaluated.

To give an example with language, look at the following sentences:

- (a) Sentence: "Shalini sat next to a friend with toys".

Meaning: The friend has toys and Shalini sat next to her.



- (b) Sentence: "Shalini sat next to a friend, with toys".

Meaning: Shalini has the toys and she sat with them next to her friend.

This sentence without the punctuation could have been interpreted in two different ways. The appropriate use of a comma specifies how the sentence has to be understood.

Let us see an expression that can be evaluated in more than one way.

- ? **Example 4:** Mallesh brought 30 marbles to the playground. Arun brought 5 bags of marbles with 4 marbles in each bag. How many marbles did Mallesh and Arun bring to the playground?

Mallesh summarized this by writing the mathematical expression —

$$30 + 5 \times 4.$$

Without knowing the context behind this expression, Purna found the value of this expression to be 140. He added 30 and 5 first, to get 35, and then multiplied 35 by 4 to get 140.

Mallesha found the value of this expression to be 50. He multiplied 5 and 4 first to get 20 and added 20 to 30 to get 50.

In this case, Mallesha is right. But why did Purna get it wrong?

Just looking at the expression $30 + 5 \times 4$, it is not clear whether we should do the addition first or multiplication.

Just as punctuation marks are used to resolve confusions in language, brackets and the notion of terms are used in mathematics to resolve confusions in evaluating expressions.

Brackets in Expressions

In the expression to find the number of marbles — $30 + 5 \times 4$ — we had to first multiply 5 and 4, and then add this product to 30. This order of operations is clarified by the use of brackets as follows:

$$30 + (5 \times 4).$$

When evaluating an expression having brackets, we need to first find the values of the expressions inside the brackets before performing other operations. So, in the above expression, we first find the value of 5×4 , and then do the addition. Thus, this expression describes the number of marbles:

$$30 + (5 \times 4) = 30 + 20 = 50.$$

Example 5: Irfan bought a pack of biscuits for ₹15 and a packet of *toor dal* for ₹56. He gave the shopkeeper ₹100. Write an expression that can help us calculate the change Irfan will get back from the shopkeeper.

Irfan spent ₹15 on a biscuit packet and ₹56 on *toor dal*. So, the total cost in rupees is $15 + 56$. He gave ₹100 to the shopkeeper. So, he should get back 100 minus the total cost. Can we write that expression as—

$$100 - 15 + 56 ?$$

Can we first subtract 15 from 100 and then add 56 to the result? We will get 141. It is absurd that he gets more money than he paid the shopkeeper!

We can use brackets in this case:

$$100 - (15 + 56).$$

Evaluating the expression within the brackets first, we get 100 minus 71, which is 29. So, Irfan will get back ₹29.

Terms in Expressions

Suppose we have the expression $30 + 5 \times 4$ without any brackets. Does it have no meaning?

When there are expressions having multiple operations, and the order of operations is not specified by the brackets, we use the notion of terms to determine the order.

Terms are the parts of an expression separated by a '+' sign. For example, in $12+7$, the terms are 12 and 7, as marked below.

$$12 + 7 = \textcircled{12} + \textcircled{7}$$

We will keep marking each term of an expression as above. Note that this way of marking the terms is not a usual practice. This will be done until you become familiar with this concept.

Now, what are the terms in $83 - 14$? We know that subtracting a number is the same as adding the inverse of the number. Recall that the inverse of a given number has the sign opposite to it. For example, the inverse of 14 is -14 , and the inverse of -14 is 14. Thus, subtracting 14 from 83 is the same as adding -14 to 83. That is,

$$83 - 14 = \textcircled{83} + \textcircled{-14}$$

Thus, the terms of the expression $83 - 14$ are 83 and -14 .

- ❓ Check if replacing subtraction by addition in this way does not change the value of the expression, by taking different examples.
- ❓ Can you explain why subtracting a number is the same as adding its inverse, using the Token Model of integers that we saw in the Class 6 textbook of mathematics?



All subtractions in an expression are converted to additions in this manner to identify the terms.

Here are some more examples of expressions and their terms:

$$-18 - 3 = \textcircled{-18} + \textcircled{-3}$$

$$6 \times 5 + 3 = \textcircled{6 \times 5} + \textcircled{3}$$

$$2 - 10 + 4 \times 6 = \textcircled{2} + \textcircled{-10} + \textcircled{4 \times 6}$$

Note that 6×5 , 4×6 are single terms as they do not have any '+' sign. In the following table, some expressions are given. Complete the table.

Expression	Expression as the sum of its terms	Terms
$13 - 2 + 6$	$(13) + (-2) + (6)$	$13, -2, 6$
$5 + 6 \times 3$	$(5) + (6 \times 3)$	
$4 + 15 - 9$	$(\quad) + (\quad) + (\quad)$	
$23 - 2 \times 4 + 16$	$(\quad) + (\quad) + (\quad)$	
$28 + 19 - 8$	$(\quad) + (\quad) + (\quad)$	

Now we will see how terms are used to determine the order of operations to find the value of an expression.

We will start with expressions having only additions (with all the subtractions suitably converted into additions).

- ❓ Does changing the order in which the terms are added give different values?

Swapping and Grouping

Let us consider a simple expression having only two terms.

- ❓ **Example 6:** Madhu is flying a drone from a terrace. The drone goes 6 m up and then 4 m down. Write an expression to show how high the final position of the drone is from the terrace.

The drone is $6 - 4 = 2$ m above the terrace. Writing it as sum of terms:

$$(6) + (-4) = (2)$$

Will the sum change if we swap the terms?

$$(-4) + (6) = (2)$$

It doesn't in this case.

We already know that swapping the terms does not change the sum when both the terms are positive numbers.

- ❓ Will this also hold when there are terms having negative numbers as well? Take some more expressions and check.

- ? Can you explain why this is happening using the Token Model of integers that we saw in the Class 6 textbook of mathematics?

Try
This

Thus, in an expression having two terms, swapping them does not change the value.

$$\text{Term 1} + \text{Term 2} = \text{Term 2} + \text{Term 1}$$

Now consider an expression having three terms: $(-7) + 10 + (-11)$. Let us add these terms in the following two different orders:

$$(-7) + 10 + (-11)$$

(adding the first two terms and then adding their sum to the third term)

$$(-7) + 10 + (-11)$$

(adding the last two terms and then adding their sum to the first term)

What do you see? The sums are the same in both cases.

Again, we know that while adding positive numbers, grouping them in any of the above two ways gives the same sum.

- ? Will this also hold when there are terms having negative numbers as well? Take some more expressions and check.
- ? Can you explain why this is happening using the Token Model of integers that we saw in the Class 6 textbook of mathematics?

Try
This

Thus, grouping the terms of an expression in either of the following ways gives the same value.

$$\text{Term 1} + \text{Term 2} + \text{Term 3} = \text{Term 1} + \text{Term 2} + \text{Term 3}$$

Let us consider the expression $(-7) + 10 + (-11)$ again. What happens when we change the order and add -7 and -11 first, and then add this sum to 10 ? Will we get the same sum as before?

We see that adding the terms of the expression $(-7) + 10 + (-11)$ in any order gives the same sum of -8 .

- ? Does adding the terms of an expression in any order give the same value? Take some more expressions and check. Consider expressions with more than 3 terms also.
- ? Can you explain why this is happening using the Token Model of integers that we saw in the Class 6 textbook of mathematics?



Thus, the addition of terms in any order gives the same value.

Therefore, in an expression having only additions, it does not matter in what order the terms are added: they all give the same value.

Now let us consider expressions having multiplication and division also, without the order of operations specified by the brackets. The values of such expressions are found by first evaluating the terms. Once all the terms are evaluated, they are added.

For example, the expression $30 + 5 \times 4$ is evaluated as follows:

$$30 + 5 \times 4 = \textcircled{30} + \textcircled{5 \times 4} = \textcircled{30} + \textcircled{20} = \textcircled{50}$$

The expression $5 \times (3 + 2) + 78 + 3$ is evaluated as follows:

$$5 \times (3 + 2) + 78 + 3 = \textcircled{5 \times (3 + 2)} + \textcircled{7 \times 8} + \textcircled{3}$$

Where $(3+2)$ is first evaluated and this sum is multiplied by 5 ($= 25$). The expression 7×8 is evaluated ($= 56$). This simplifies to $25 + 56 + 3 = 84$.

- ? Manasa is adding a long list of numbers. It took her five minutes to add them all and she got the answer 11749. Then she realised that she had forgotten to include the fourth number 9055. Does she have to start all over again?

1342
774
8611
9055
1022

In mathematics we use the phrase **commutative property** of **addition** instead of saying “swapping terms does not change the sum”. Similarly, “grouping does not change the sum” is called the **associative property of addition**.

Swapping the Order of Things in Everyday Life

- ? Manasa is going outside to play. Her mother says, “Wear your hat and shoes!” Which one should she wear first? She can wear her hat first and then her shoes. Or she can wear her shoes first and then her hat.



Manasa will look exactly the same in both cases. Imagine a different situation: Manasa’s mother says “Wear your socks and shoes!” Now the



order matters. She should wear socks and then shoes. If she wears shoes and then socks, Manasa will feel very uncomfortable and look very different.

More Expressions and Their Terms

- Example 7:** Amu, Charan, Madhu, and John went to a hotel and ordered four dosas. Each dosa cost ₹23, and they wish to thank the waiter by tipping ₹5. Write an expression describing the total cost.

$$\text{Cost of 4 dosas} = 4 \times 23$$

Can the total amount with tip be written as $4 \times 23 + 5$? Evaluating it, we get

$$4 \times 23 + 5 = (4 \times 23) + 5 = 92 + 5 = 97$$

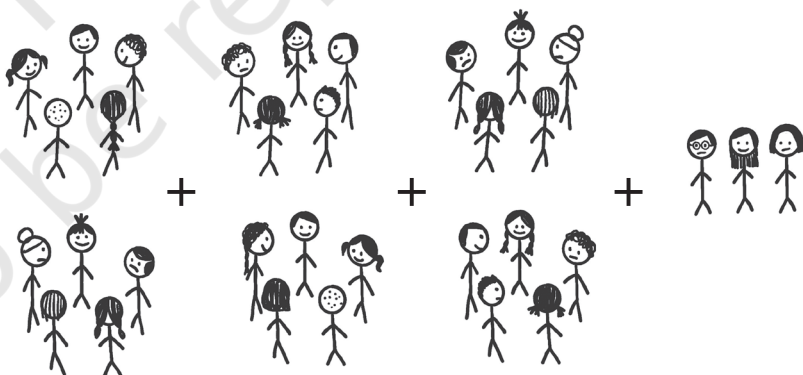
Thus, $4 \times 23 + 5$ is a correct way of writing the expression.

- Example 8:** Children in a class are playing “Fire in the mountain, run, run, run!”. Whenever the teacher calls out a number, students are supposed to arrange themselves in groups of that number. Whoever is not part of the announced group size, is out.

Ruby wanted to rest and sat on one side. The other 33 students were playing the game in the class.

The teacher called out ‘5’. Once children settled, Ruby wrote $6 \times 5 + 3$

(understood as 3 more than 6×5)



- Think and discuss why she wrote this.

The expression written as a sum of terms is—

$$(6 \times 5) + 3$$

- ? For each of the cases below, write the expression and identify its terms:
 If the teacher had called out '4', Ruby would write _____
 If the teacher had called out '7', Ruby would write _____
 Write expressions like the above for your class size.

- ? **Example 9:** Raghu bought 100 kg of rice from the wholesale market and packed them into 2 kg packets. He already had four 2 kg packets. Write an expression for the number of 2 kg packets of rice he has now and identify the terms.

He had 4 packets. The number of new 2 kg packets of rice is $100 \div 2$, which we also write as $\frac{100}{2}$.

The number of 2 kg packets he has now is $4 + \frac{100}{2}$. The terms are—

$$\left(4 \right) + \left(\frac{100}{2} \right).$$

- ? **Example 10:** Kannan has to pay ₹432 to a shopkeeper using coins of ₹1 and ₹5, and notes of ₹10, ₹20, ₹50 and ₹100. How can he do it?

There is more than one possibility. For example,

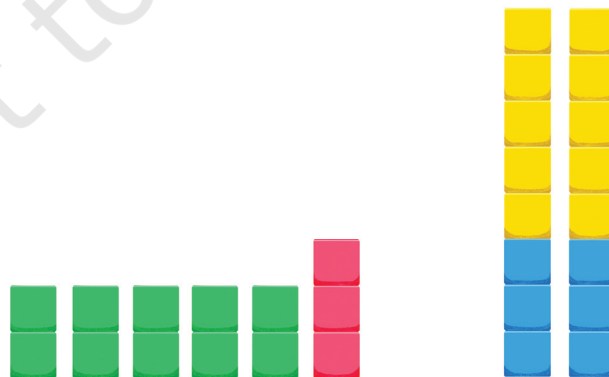
$$432 = 4 \times 100 + 1 \times 20 + 1 \times 10 + 2 \times 1$$

Meaning: 4 notes of ₹100, 1 note of ₹20, 1 note of ₹10 and 2 notes of ₹1

$$432 = 8 \times 50 + 1 \times 10 + 4 \times 5 + 2 \times 1$$

Meaning: 8 notes of ₹50, 1 note of ₹10, 4 notes of ₹5 and 2 notes of ₹1

- ? Identify the terms in the two expressions above.
 ? Can you think of some more ways of giving ₹432 to someone?
 ? **Example 11:** Here are two pictures. Which of these two arrangements matches with the expression $5 \times 2 + 3$?



Let us write this expression as a sum of terms.

$$(5 \times 2) + (3) = (10) + (3) = (13)$$

This expression $5 \times 2 + 3$ can be understood as 3 more than 5×2 , which describes the arrangement on the left.

- ❓ What is the expression for the arrangement in the right making use of the number of yellow and blue squares?

Do you recall the use of brackets? We need to use brackets for this.

$$2 \times (5 + 3)$$

Notice that this arrangement can also be described using—

$$5 + 3 + 5 + 3$$

OR

$$5 \times 2 + 3 \times 2$$

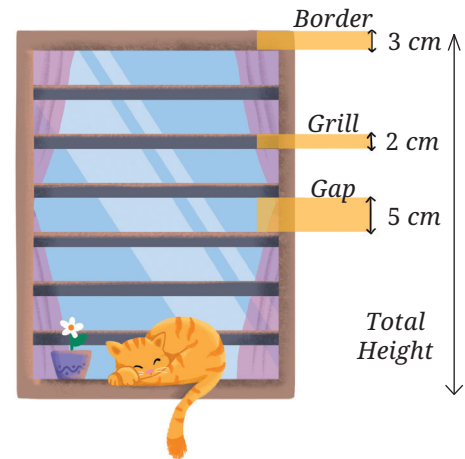
❓ Figure it Out

- Find the values of the following expressions by writing the terms in each case.

(a) $28 - 7 + 8$	(b) $39 - 2 \times 6 + 11$
(c) $40 - 10 + 10 + 10$	(d) $48 - 10 \times 2 + 16 \div 2$
(e) $6 \times 3 - 4 \times 8 \times 5$	
- Write a story/situation for each of the following expressions and find their values.

(a) $89 + 21 - 10$	(b) $5 \times 12 - 6$
(c) $4 \times 9 + 2 \times 6$	
- For each of the following situations, write the expression describing the situation, identify its terms and find the value of the expression.
 - Queen Alia gave 100 gold coins to Princess Elsa and 100 gold coins to Princess Anna last year. Princess Elsa used the coins to start a business and doubled her coins. Princess Anna bought jewellery and has only half of the coins left. Write an expression describing how many gold coins Princess Elsa and Princess Anna together have.
 - A metro train ticket between two stations is ₹40 for an adult and ₹20 for a child. What is the total cost of tickets:
 - for four adults and three children?
 - for two groups having three adults each?

- (c) Find the total height of the window by writing an expression describing the relationship among the measurements shown in the picture.



Removing Brackets—I

Let us find the value of this expression,

$$200 - (40 + 3).$$

We first evaluate the expression inside the bracket to 43 and then subtract it from 200. But it is simpler to first subtract 40 from 200:

$$200 - 40 = 160.$$

And then subtract 3 from 160:

$$160 - 3 = 157.$$

What we did here was $200 - 40 - 3$. Notice, that we did not do

$$200 - 40 + 3.$$

So,

$$200 - (40 + 3) = 200 - 40 - 3.$$

Example 12: We also saw this earlier in the case of Irfan purchasing a biscuit packet (₹15) and a *toor dal* packet (₹56). When he paid ₹100, the change he gets in rupees is:

$$100 - (15 + 56) = 29.$$

The change could also have been calculated as follows:

- (a) First subtract the cost of the biscuit packet (15) from 100:

$$100 - 15 = 85.$$

This is the amount the shopkeeper owes Irfan if he had purchased only the biscuits. As he has purchased *toor dal* also, its cost is taken from this remaining amount of 85.

- (b) So, to find the change, we need to subtract the cost of *toor dal* from 85.

$$85 - 56 = 29.$$

What we have done here is $100 - 15 - 56$. So,

$$100 - (15 + 56) = 100 - 15 - 56.$$

Notice how upon **removing the brackets preceded by a negative sign**, the signs of the terms inside the brackets change. Observe

the signs of 40 and 3 in the first example, and that of 15 and 56 in the second.

Example 13: Consider the expression $500 - (250 - 100)$. Is it possible to write this expression without the brackets?

To evaluate this expression, we need to subtract $250 - 100 = 150$ from 500:

$$500 - (250 - 100) = 500 - 150 = 350.$$

If we were to directly subtract 250 from 500, then we would have subtracted 100 more than what we needed to. So, we should add back that 100 to $500 - 250$ to make the expression take the same value as $500 - (250 - 100)$. This sequence of operations is $500 - 250 + 100$. Thus,

$$500 - (250 - 100) = 500 - 250 + 100.$$

Check that $500 - (250 - 100)$ is not equal to $500 - 250 - 100$.

Notice again that **when the brackets preceded by a negative sign are removed, the signs of the terms inside the brackets change**. In this case, the signs of 250 and -100 change to -250 and 100.

Example 14: Hira has a rare coin collection. She has 28 coins in one bag and 35 coins in another. She gifts her friend 10 coins from the second bag. Write an expression for the number of coins left with Hira.

This can be expressed by $28 + (35 - 10)$.

We know that this is the same as $28 + (35 + (-10))$. Since the terms can be added in any order, this expression can simply be written as $28 + 35 + (-10)$, or $28 + 35 - 10$. Thus,

$$28 + (35 - 10) = 28 + 35 - 10 = 53.$$

When the brackets are NOT preceded by a negative sign, the terms within them do not change their signs upon removing the brackets. Notice the sign of the terms 35 and -10 in the above expression.



Rather than simply remembering rules for when to change the sign and when not to, you can figure it out for yourself by thinking about the meanings of the expressions.

Tinker the Terms I

What happens to the value of an expression if we increase or decrease the value of one of its terms?

Some expressions are given in following three columns. In each column, one or more terms are changed from the first expression. Go through the example (in the first column) and fill the blanks, doing as little computation as possible.

? $53 + (-16) = 37$	$53 + (-16) = 37$	$-87 + (-16) = \bigcirc$
$54 + (-16) = 38$ 54 is one more than 53, so the value will be 1 more than 37.	$52 + (-16) = \bigcirc$ 52 is one less than 53, so the value will be 1 less than 37.	$-88 + (-15) = \bigcirc$
$53 + (-15) = \bigcirc$ Is -15 one more or one less than -16?	$53 + (-17) = \bigcirc$ Is -17 one more or one less than -16?	$-86 + (-18) = \bigcirc$
		$-97 + (-26) = \bigcirc$

? Figure it Out

- Fill in the blanks with numbers, and boxes with operation signs such that the expressions on both sides are equal.
 - $24 + (6 - 4) = 24 + 6 \square$
 - $38 + (\square \square) = 38 + 9 - 4$
 - $24 - (6 + 4) = 24 \square 6 - 4$
 - $24 - 6 - 4 = 24 - 6 \square$
 - $27 - (8 + 3) = 27 \square 8 \square 3$
 - $27 - (\square \square) = 27 - 8 + 3$
- Remove the brackets and write the expression having the same value.

(a) $14 + (12 + 10)$	(b) $14 - (12 + 10)$
(c) $14 + (12 - 10)$	(d) $14 - (12 - 10)$
(e) $-14 + 12 - 10$	(f) $14 - (-12 - 10)$
- Find the values of the following expressions. For each pair, first try to guess whether they have the same value. When are the two expressions equal?
 - $(6 + 10) - 2$ and $6 + (10 - 2)$
 - $16 - (8 - 3)$ and $(16 - 8) - 3$
 - $27 - (18 + 4)$ and $27 + (-18 - 4)$
- In each of the sets of expressions below, identify those that have the same value. Do not evaluate them, but rather use your understanding of terms.

- (a) $319 + 537$, $319 - 537$, $-537 + 319$, $537 - 319$
- (b) $87 + 46 - 109$, $87 + 46 - 109$, $87 + 46 - 109$, $87 - 46 + 109$, $87 - (46 + 109)$, $(87 - 46) + 109$
5. Add brackets at appropriate places in the expressions such that they lead to the values indicated.
- (a) $34 - 9 + 12 = 13$
- (b) $56 - 14 - 8 = 34$
- (c) $-22 - 12 + 10 + 22 = -22$
6. Using only reasoning of how terms change their values, fill the blanks to make the expressions on either side of the equality (=) equal.
- (a) $423 + \underline{\hspace{2cm}} = 419 + \underline{\hspace{2cm}}$
- (b) $207 - 68 = 210 - \underline{\hspace{2cm}}$
7. Using the numbers 2, 3 and 5, and the operators '+' and '-', and brackets, as necessary, generate expressions to give as many different values as possible. For example, $2 - 3 + 5 = 4$ and $3 - (5 - 2) = 0$.
8. Whenever Jasoda has to subtract 9 from a number, she subtracts 10 and adds 1 to it. For example, $36 - 9 = 26 + 1$.
- (a) Do you think she always gets the correct answer? Why?
- (b) Can you think of other similar strategies? Give some examples.
9. Consider the two expressions: a) $73 - 14 + 1$, b) $73 - 14 - 1$. For each of these expressions, identify the expressions from the following collection that are equal to it.
- (a) $73 - (14 + 1)$ b) $73 - (14 - 1)$
- (c) $73 + (-14 + 1)$ d) $73 + (-14 - 1)$



Removing Brackets—II

- Example 15:** Lhamo and Norbu went to a hotel. Each of them ordered a vegetable cutlet and a *rasgulla*. A vegetable cutlet costs ₹43 and a *rasgulla* costs ₹24. Write an expression for the amount they will have to pay.

As each of them had one vegetable cutlet and one *rasagulla*, each of their shares can be represented by $43 + 24$.

- ?** What about the total amount they have to pay? Can it be described by the expression: $2 \times 43 + 24$?

Writing it as sum of terms gives:

$$(2 \times 43) + (24)$$

This expression means 24 more than 2×43 . But, we want an expression which means twice or double of $43 + 24$.

We can make use of brackets to write such an expression:

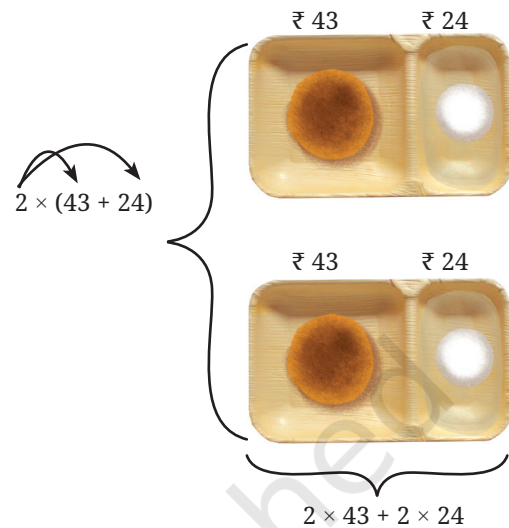
$$2 \times (43 + 24).$$

So, we can say that together they have to pay $2 \times (43 + 24)$. This is also the same as paying for two vegetable cutlets and two *rasgullas*:

$$2 \times 43 + 2 \times 24.$$

Therefore,

$$2 \times (43 + 24) = 2 \times 43 + 2 \times 24.$$



? If another friend, Sangmu, joins them and orders the same items, what will be the expression for the total amount to be paid?

? **Example 16:** In the Republic Day parade, there are boy scouts and girl guides marching together. The scouts march in 4 rows with 5 scouts in each row. The guides march in 3 rows with 5 guides in each row (see the figure below). How many scouts and guides are marching in this parade?

The number of boy scouts marching is 4×5 . The number of girl guides marching is 3×5 .

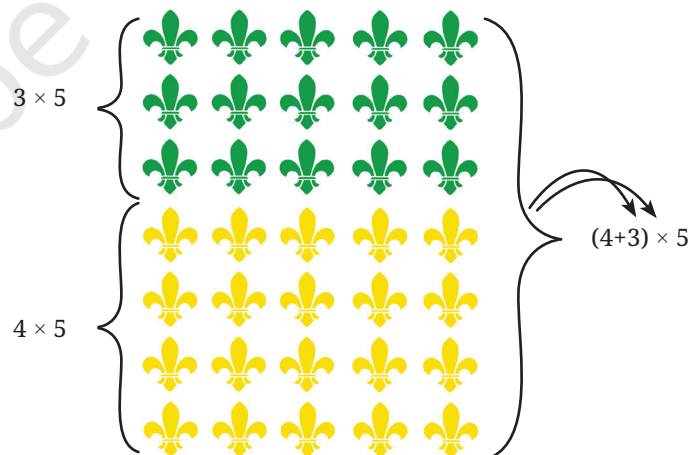
The total number of scouts and guides will be $4 \times 5 + 3 \times 5$.

This can also be found by first finding the total number of rows, i.e., $4 + 3$, and then multiplying their sum by the number of children in each row. Thus, the number of boys and girls can be found by $(4 + 3) \times 5$.

Therefore, $4 \times 5 + 3 \times 5$
 $= (4 + 3) \times 5$.

Computing these expressions, we get

$$4 \times 5 + 3 \times 5 = (4 \times 5) + (3 \times 5) = 20 + 15 = 35$$



$$(4 + 3) \times 5 = 7 \times 5 = 35$$

❓ $5 \times 4 + 3 \neq 5 \times (4 + 3)$. Can you explain why?

$$\text{Is } 5 \times (4 + 3) = 5 \times (3 + 4) = (3 + 4) \times 5?$$

The observations that we have made in the previous two examples can be seen in a general way as follows.

Consider $10 \times 98 + 3 \times 98$. This means taking the sum of 10 times 98 and 3 times 98.

$$\underbrace{98 + 98 + 98 + 98 + 98 + 98 + 98 + 98 + 98 + 98}_{10 \text{ times}} + \underbrace{98 + 98 + 98}_{3 \text{ times}}$$

Clearly, this is the same as $10 + 3 = 13$ times 98. Thus,

$$10 \times 98 + 3 \times 98 = (10 + 3) \times 98.$$

Writing this equality the other way, we get

$$(10 + 3) 98 = 10 \times 98 + 3 \times 98.$$

Swapping the numbers in the products above, this property can be seen in the following form:

$$98 \times 10 + 98 \times 3 = 98 (10 + 3), \text{ and}$$

$$98 (10 + 3) = 98 \times 10 + 98 \times 3.$$

Similarly, let us consider the expression $14 \times 10 - 6 \times 10$. This means subtracting 6 times 10 from 14 times 10.

$$\underbrace{10 + 10 + 10 + 10 + 10 + 10 + 10 + 10 + 10 + 10 + 10 + 10 + 10 + 10}_{14 \text{ times}} - \underbrace{10 + 10 + 10 + 10 + 10 + 10}_{6 \text{ times}}$$

Clearly, this is $14 - 6 = 8$ times 10. Thus,

$$14 \times 10 - 6 \times 10 = (14 - 6) \times 10,$$

or

$$(14 - 6) \times 10 = 14 \times 10 - 6 \times 10$$

This property can be nicely summed up as follows:

The multiple of a sum (difference) is the same as the sum (difference) of the multiples.

Tinker the Terms II

Let us understand what happens when we change the numbers occurring in a product.

Example 17: Given $53 \times 18 = 954$. Find out 63×18 .

As 63×18 means 63 times 18,

$$\begin{aligned} 63 \times 18 &= (53 + 10) \times 18 \\ &= 53 \times 18 + 10 \times 18 \\ &= 954 + 180 \\ &= 1134. \end{aligned}$$

Example 18: Find an effective way of evaluating 97×25 .

97×25 means 97 times 25.

We can write it as $(100 - 3) \times 25$

We know that this is the same as the difference of 100 times 25 and 3 times 25:

$$97 \times 25 = 100 \times 25 - 3 \times 25$$

Find this value.

Use this method to find the following products:

- (a) 95×8
- (b) 104×15
- (c) 49×50

Is this quicker than the multiplication procedure you use generally?

Which other products might be quicker to find like the ones above?



Figure it Out

1. Fill in the blanks with numbers, and boxes by signs, so that the expressions on both sides are equal.

- (a) $3 \times (6 + 7) = 3 \times 6 + 3 \times 7$
- (b) $(8 + 3) \times 4 = 8 \times 4 + 3 \times 4$
- (c) $3 \times (5 + 8) = 3 \times 5 \boxed{} 3 \times \underline{\hspace{1cm}}$
- (d) $(9 + 2) \times 4 = 9 \times 4 \boxed{} 2 \times \underline{\hspace{1cm}}$
- (e) $3 \times (\underline{\hspace{1cm}} + 4) = 3 \underline{\hspace{1cm}} + \underline{\hspace{1cm}}$
- (f) $(\underline{\hspace{1cm}} + 6) \times 4 = 13 \times 4 + \underline{\hspace{1cm}}$
- (g) $3 \times (\underline{\hspace{1cm}} + \underline{\hspace{1cm}}) = 3 \times 5 + 3 \times 2$
- (h) $(\underline{\hspace{1cm}} + \underline{\hspace{1cm}}) \times \underline{\hspace{1cm}} = 2 \times 4 + 3 \times 4$
- (i) $5 \times (9 - 2) = 5 \times 9 - 5 \times \underline{\hspace{1cm}}$
- (j) $(5 - 2) \times 7 = 5 \times 7 - 2 \times \underline{\hspace{1cm}}$
- (k) $5 \times (8 - 3) = 5 \times 8 \boxed{} 5 \times \underline{\hspace{1cm}}$
- (l) $(8 - 3) \times 7 = 8 \times 7 \boxed{} 3 \times 7$

(m) $5 \times (12 - \underline{\quad}) = \underline{\quad} \boxed{\quad} 5 \times \underline{\quad}$

(n) $(15 - \underline{\quad}) \times 7 = \underline{\quad} \boxed{\quad} 6 \times 7$

(o) $5 \times (\underline{\quad} - \underline{\quad}) = 5 \times 9 - 5 \times 4$

(p) $(\underline{\quad} - \underline{\quad}) \times \underline{\quad} = 17 \times 7 - 9 \times 7$

2. In the boxes below, fill '<', '>' or '=' after analysing the expressions on the LHS and RHS. Use reasoning and understanding of terms and brackets to figure this out and not by evaluating the expressions.

(a) $(8 - 3) \times 29$ $\boxed{\quad}$ $(3 - 8) \times 29$

(b) $15 + 9 \times 18$ $\boxed{\quad}$ $(15 + 9) \times 18$

(c) $23 \times (17 - 9)$ $\boxed{\quad}$ $23 \times 17 + 23 \times 9$

(d) $(34 - 28) \times 42$ $\boxed{\quad}$ $34 \times 42 - 28 \times 42$

3. Here is one way to make 14: $\underline{2} \times (\underline{1} + \underline{6}) = 14$. Are there other ways of getting 14? Fill them out below:

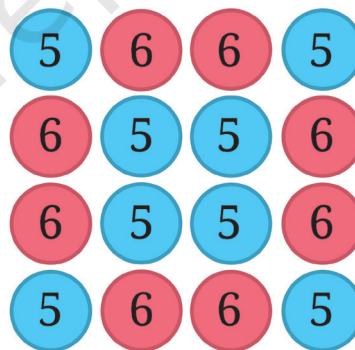
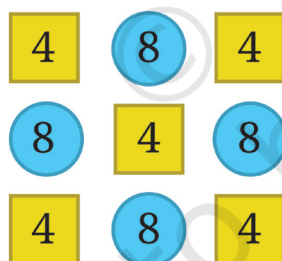
(a) $\underline{\quad} \times (\underline{\quad} + \underline{\quad}) = 14$

(b) $\underline{\quad} \times (\underline{\quad} + \underline{\quad}) = 14$

(c) $\underline{\quad} \times (\underline{\quad} + \underline{\quad}) = 14$

(d) $\underline{\quad} \times (\underline{\quad} + \underline{\quad}) = 14$

4. Find out the sum of the numbers given in each picture below in at least two different ways. Describe how you solved it through expressions.



? Figure it Out

1. Read the situations given below. Write appropriate expressions for each of them and find their values.
- (a) The district market in Begur operates on all seven days of a week. Rahim supplies 9 kg of mangoes each day from his orchard and Shyam supplies 11 kg of mangoes each day from his orchard to this market. Find the amount of mangoes supplied by them in a week to the local district market.

- (b) Binu earns ₹20,000 per month. She spends ₹5,000 on rent, ₹5,000 on food, and ₹2,000 on other expenses every month. What is the amount Binu will save by the end of a year?
- (c) During the daytime a snail climbs 3 cm up a post, and during the night while asleep, accidentally slips down by 2 cm. The post is 10 cm high, and a delicious treat is on its top. In how many days will the snail get the treat?
2. Melvin reads a two-page story every day except on Tuesdays and Saturdays. How many stories would he complete reading in 8 weeks? Which of the expressions below describes this scenario?
- (a) $5 \times 2 \times 8$
 (b) $(7 - 2) \times 8$
 (c) 8×7
 (d) $7 \times 2 \times 8$
 (e) $7 \times 5 - 2$
 (f) $(7 + 2) \times 8$
 (g) $7 \times 8 - 2 \times 8$
 (h) $(7 - 5) \times 8$
3. Find different ways of evaluating the following expressions:
- (a) $1 - 2 + 3 - 4 + 5 - 6 + 7 - 8 + 9 - 10$
 (b) $1 - 1 + 1 - 1 + 1 - 1 + 1 - 1 + 1 - 1$
4. Compare the following pairs of expressions using '<', '>' or '=' or by reasoning.
- | | | |
|---------------------------|----------------------|--------------------------------|
| (a) $49 - 7 + 8$ | <input type="text"/> | $49 - 7 + 8$ |
| (b) $83 \times 42 - 18$ | <input type="text"/> | $83 \times 40 - 18$ |
| (c) $145 - 17 \times 8$ | <input type="text"/> | $145 - 17 \times 6$ |
| (d) $23 \times 48 - 35$ | <input type="text"/> | $23 \times (48 - 35)$ |
| (e) $(16 - 11) \times 12$ | <input type="text"/> | $-11 \times 12 + 16 \times 12$ |
| (f) $(76 - 53) \times 88$ | <input type="text"/> | $88 \times (53 - 76)$ |
| (g) $25 \times (42 + 16)$ | <input type="text"/> | $25 \times (43 + 15)$ |
| (h) $36 \times (28 - 16)$ | <input type="text"/> | $35 \times (27 - 15)$ |

5. Identify which of the following expressions are equal to the given expression without computation. You may rewrite the expressions using terms or removing brackets. There can be more than one expression which is equal to the given expression.

(a) $83 - 37 - 12$

(i) $84 - 38 - 12$

(ii) $84 - (37 + 12)$

(iii) $83 - 38 - 13$

(iv) $-37 + 83 - 12$

(b) $93 + 37 \times 44 + 76$

(i) $37 + 93 \times 44 + 76$

(ii) $93 + 37 \times 76 + 44$

(iii) $(93 + 37) \times (44 + 76)$

(iv) $37 \times 44 + 93 + 76$

5. Choose a number and create ten different expressions having that value.

SUMMARY

- We have been reading and evaluating simple expressions for quite some time now. Here we started by revising the meaning of some simple expressions and their values.
- We learnt how to compare certain expressions through reasoning instead of bluntly evaluating them.
- To help read and evaluate complex expressions without confusion, we use terms and brackets.
- When an expression is written as a sum of terms, changing the order of the terms or grouping the terms does not change the value of the expression. This is because the “commutative property of addition” and the “associative property of addition”, respectively.
- To evaluate expressions within brackets, we saw that when we remove brackets preceded by a negative sign, the terms within the bracket change their sign.
- We also learnt about the “distributive property” — multiplying a number with an expression inside brackets is equal to the multiplying the number with each term in the bracket.



Using three 3's along with the four operations (addition, subtraction, multiplication, and division) and brackets as needed we can create several expressions. For example, $(3 + 3)/3 = 2$, $3 + 3 - 3 = 3$, $3 \times 3 + 3 = 12$, and so on.

Using four 4's, create expressions to get all values from 1 to 20.

Using the numbers 1, 2, 3, 4, and 5 exactly once in any order get as many values as possible between -10 and $+10$.

Using the numbers 0 to 9 exactly once in any order, make an expression with a value 100.

What other similar interesting questions can you ask?

